

Data Availability Statement

The public benchmarks used in this study are available from their original providers: BOP Linemod and T-LESS (BOP benchmark), SCARED (endoscopic stereo), real-colon FEM, EndoN-eRF, and BlendedMVS. The FoundationPose and Endo3R implementations compared against are publicly released by their respective teams; we use our Windows port of FoundationPose for the full-benchmark comparison.

The clinical CT-to-bronchoscopy data are de-identified patient records provided by Henan Provincial People’s Hospital and are not publicly available owing to institutional data-governance restrictions; access may be requested from the hospital subject to its data-governance process.

All evaluation scripts, the Windows build patches for FoundationPose/Endo3R, and the result figures across every dataset are released by the authors to support reproducibility.